Project Milestone 4

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2026-10-21

Load packages and datasets

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
d3_instLevel = read_csv("../clean_data/d3_instLevel_ds.csv")
d3_sports = read_csv("../clean_data/d3_sports_ds.csv")
d3_coach = read_csv("../clean_data/d3_coach_ds.csv")

d3_sports = d3_sports %>%
  mutate(
    institution_name = case_when(
      str_starts(institution_name, "Linfield") ~ "Linfield University",
      .default = institution_name
    )
  )

d3_instLevel = d3_instLevel %>%
  mutate(
    institution_name = case_when(
      str_starts(institution_name, "Linfield") ~ "Linfield University",
      .default = institution_name
    )
  )

d3_coach = d3_coach %>%
  mutate(
    institution_name = case_when(
      str_starts(institution_name, "Linfield") ~ "Linfield University",
      .default = institution_name
    )
  )

nwc = c(
  "Linfield University", "George Fox University", "Whitworth University", "Whitman College", "Pacific U
)
```

```r
d3_sports %>%
  filter(str_starts(institution_name, "Linfield"))
```

```
## # A tibble: 442 x 12
##     year unitid institution_name    sport       athlete_gender n_athletes op_exp
##    <dbl>  <dbl> <chr>              <chr>       <chr>                <dbl>  <dbl>
##  1  2005 209065 Linfield University Men's Base~ Men                    48  48311
##  2  2005 209065 Linfield University Men's Bask~ Men                    27  54547
##  3  2005 209065 Linfield University Men's Foot~ Men                   124 106755
##  4  2005 209065 Linfield University Men's Golf  Men                    13  13235
##  5  2005 209065 Linfield University Men's Socc~ Men                    26  22585
##  6  2005 209065 Linfield University Men's Swim~ Men                    17  14649
##  7  2005 209065 Linfield University Men's Tenn~ Men                    16  22678
##  8  2005 209065 Linfield University Men's Trac~ Men                    34   1456
##  9  2005 209065 Linfield University Men's Trac~ Men                    37  13105
## 10  2005 209065 Linfield University Men's Trac~ Men                    11   3559
## # i 432 more rows
## # i 5 more variables: op_exp_per_athlete <dbl>, expense <dbl>, revenue <dbl>,
## #   total_hc <dbl>, total_ac <dbl>
```

EDA plots

Plot 1: Operating Expenses vs Number of Athletes

- Geometries: Points, ABline, text
- Aesthetics: x = number of athletes, y = operating expenses, color = type of school

```r
selected_sport = "Men's Football"
selected_year = 2025

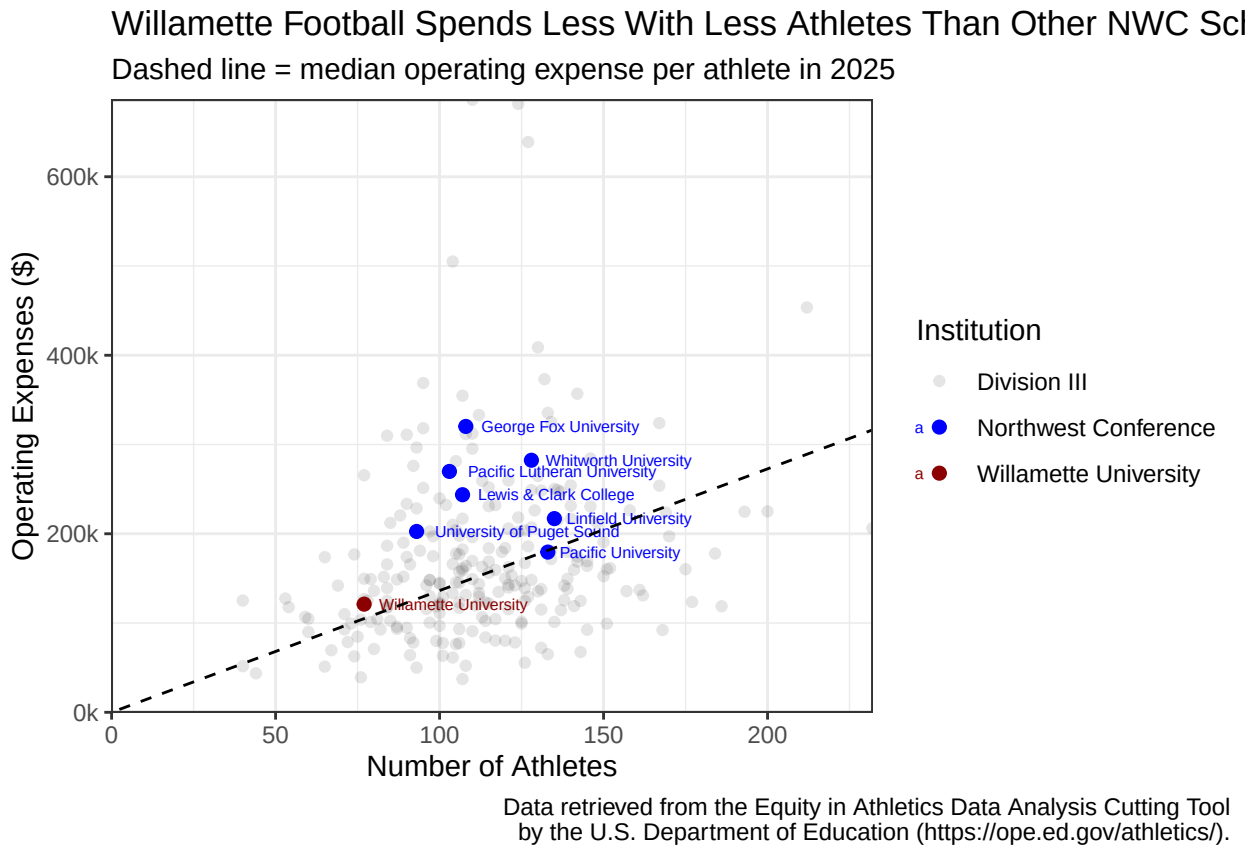
title = paste0("Divison III ", selected_sport, " Operating Expenses vs Number of Athletes in ", as.nume

ds = d3_sports %>%
  filter(sport == selected_sport, year == selected_year) %>%
  mutate(
    color = case_when(
      institution_name %in% nwc ~ "Northwest Conference",
      institution_name == "Willamette University" ~ "Willamette University",
      .default = "Division III"
    )
  )

colorpal = c(
  "Willamette University" = "darkred",
  "Northwest Conference" = "blue",
  "Division III" = "black"
)


solid = ds %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University")
```

```r
ds %>%
  filter(!(institution_name %in% nwc), !(institution_name == "Willamette University")) %>%
  ggplot(aes(x = n_athletes, y = op_exp, color = color)) +
  geom_point(alpha = .1) +
  geom_point(data = solid, aes(x = n_athletes, y = op_exp, color = color), size = 2) +
  geom_text(data = solid, aes(label = institution_name), hjust = -.1, size = 2) +
  geom_abline(intercept = 0, slope = median(ds$op_exp, na.rm = TRUE)/median(ds$n_athletes), color = "bla
  scale_color_manual(values = colorpal) +
  scale_x_continuous(limits = c(0, max(ds$n_athletes)), expand = c(0, 0)) +
  scale_y_continuous(labels = function (x) paste0(x/1000, "k"), limits = c(0, max(ds$op_exp)), expand =
  theme_bw() +
  theme(
    plot.caption.position = "plot"
  ) +
  labs(
    title = "Willamette Football Spends Less With Less Athletes Than Other NWC Schools",
    subtitle = "Dashed line = median operating expense per athlete in 2025",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
    x = "Number of Athletes",
    y = "Operating Expenses ($)",
    color = "Institution"
  )
```



This is our main point, that Willamette spends less than other schools, which leads us with less opportunities

for growth. By presenting this graphic which can show multiple different sports and years, we can show that this finding is consistent and change needs to be made.

Plot 2: Operating Expenses per Athlete by Year

- Geometries: Points, smooth line
- Aesthetics: x = year, y = operating expenses per athlete, color = type of school

```
selected_sport = "Men's Football"

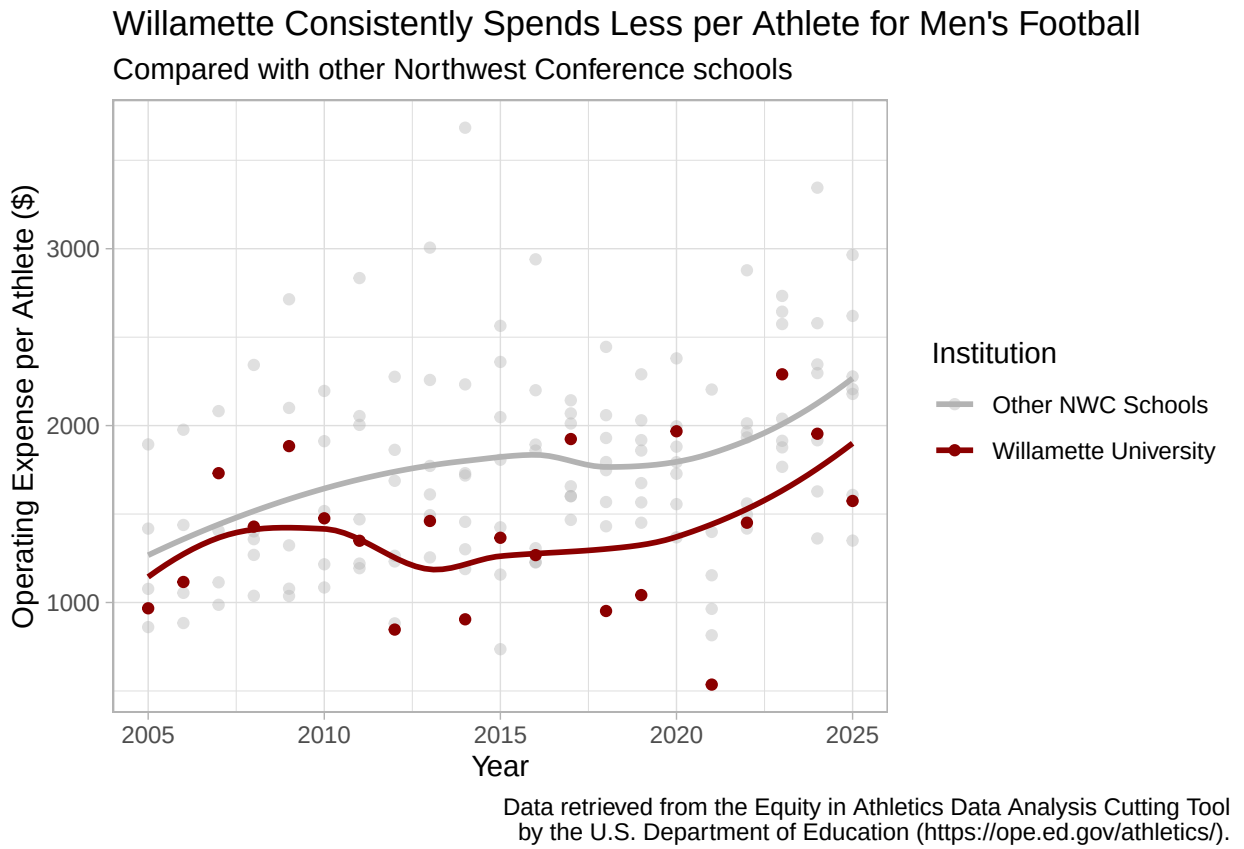
title = paste0("Willamette Consistently Spends Less per Athlete for ", selected_sport)

points = d3_sports %>%
  filter(sport == selected_sport, institution_name %in% nwc)

willamette = d3_sports %>%
  filter(sport == selected_sport, institution_name == "Willamette University")



points %>%
  ggplot(aes(x = year, y = op_exp_per_athlete)) +
  geom_point(alpha = .4, aes(color = "Other NWC Schools")) +
  geom_smooth(se = F, alpha = .5, aes(color = "Other NWC Schools")) +
  geom_point(data = willamette, aes(x = year, y = op_exp_per_athlete, color = "Willamette University")) +
  geom_smooth(data = willamette, se = F, aes(color = "Willamette University")) +
  theme_light() +
  theme(
    plot.caption.position = "plot"
  ) +
  scale_color_manual(
    values = c(
      "Willamette University" = "darkred",
      "Other NWC Schools" = "grey70"
    )
  ) +
  labs(
    title = title,
    subtitle = "Compared with other Northwest Conference schools",
    x = "Year",
    y = "Operating Expense per Athlete ($)",
    color = "Institution",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



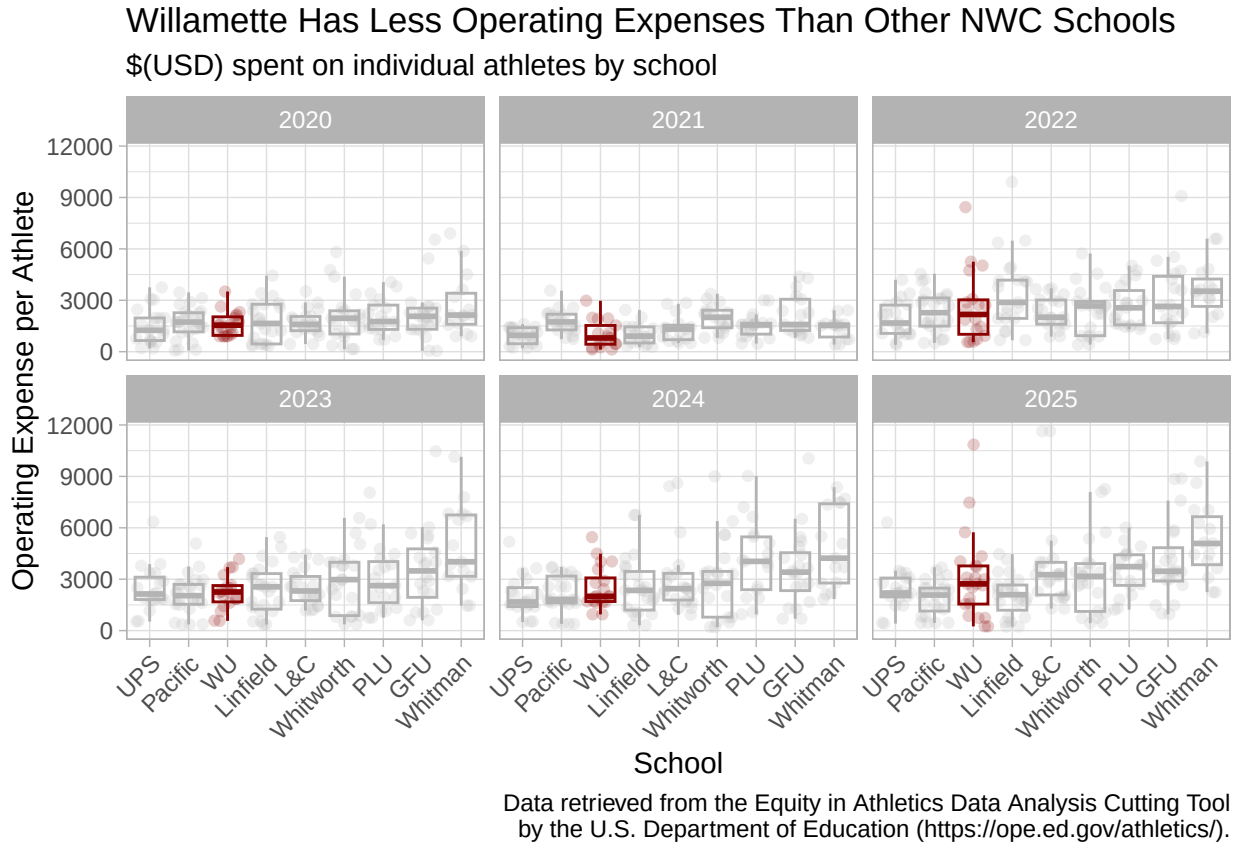
This is our main point, that Willamette spends less than other schools, which leads us with less opportunities for growth. By presenting this graphic which can show multiple different sports over time, we can show that this finding is consistent and change needs to be made.

Plot 3: Operating Expenses per Athlete by School

- Geometries: Points, boxplot
- Aesthetics: x = school, color = type of school

```
d3_sports %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University", year >= 2020) %>%
  ggplot(aes(x = fct_reorder(institution_name, op_exp_per_athlete, .fun = median), y = op_exp_per_athle
  geom_point(aes(color = institution_name == "Willamette University",
        group = institution_name), alpha = .2, position = "jitter") +
  geom_boxplot(aes(color = institution_name == "Willamette University", group = institution_name), alpha
  facet_wrap(~year) +
  theme_light() +
  theme(
    plot.caption.position = "plot",
    axis.text.x = element_text(angle = 45, hjust = 1)
  )  +
  scale_x_discrete(
    labels = c(
      "Willamette University" = "WU", "University of Puget Sound" = "UPS", "Lewis & Clark College" = "L
    )
```

```
  ) +
  scale_color_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    guide = "none"
  ) +
  labs(
    x = "School",
    y = "Operating Expense per Athlete",
    title = "Willamette Has Less Operating Expenses Than Other NWC Schools",
    subtitle = "$(USD) spent on individual athletes by school",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



This is our main point, that Willamette spends less than other schools, which leads us with less opportunities for growth. By presenting this graphic which shows data for all sports at once across multiple different years, we can show that this finding is consistent and change needs to be made.

```
library(tidytext)
```

```
school_labels = c(
  "Willamette University" = "WU", "University of Puget Sound" = "UPS",
  "Lewis & Clark College" = "L&C", "Pacific Lutheran University" = "PLU",
  "Pacific University" = "Pacific", "Whitworth University" = "Whitworth",
  "George Fox University" = "GFU", "Linfield University" = "Linfield",
  "Whitman College" = "Whitman"
```

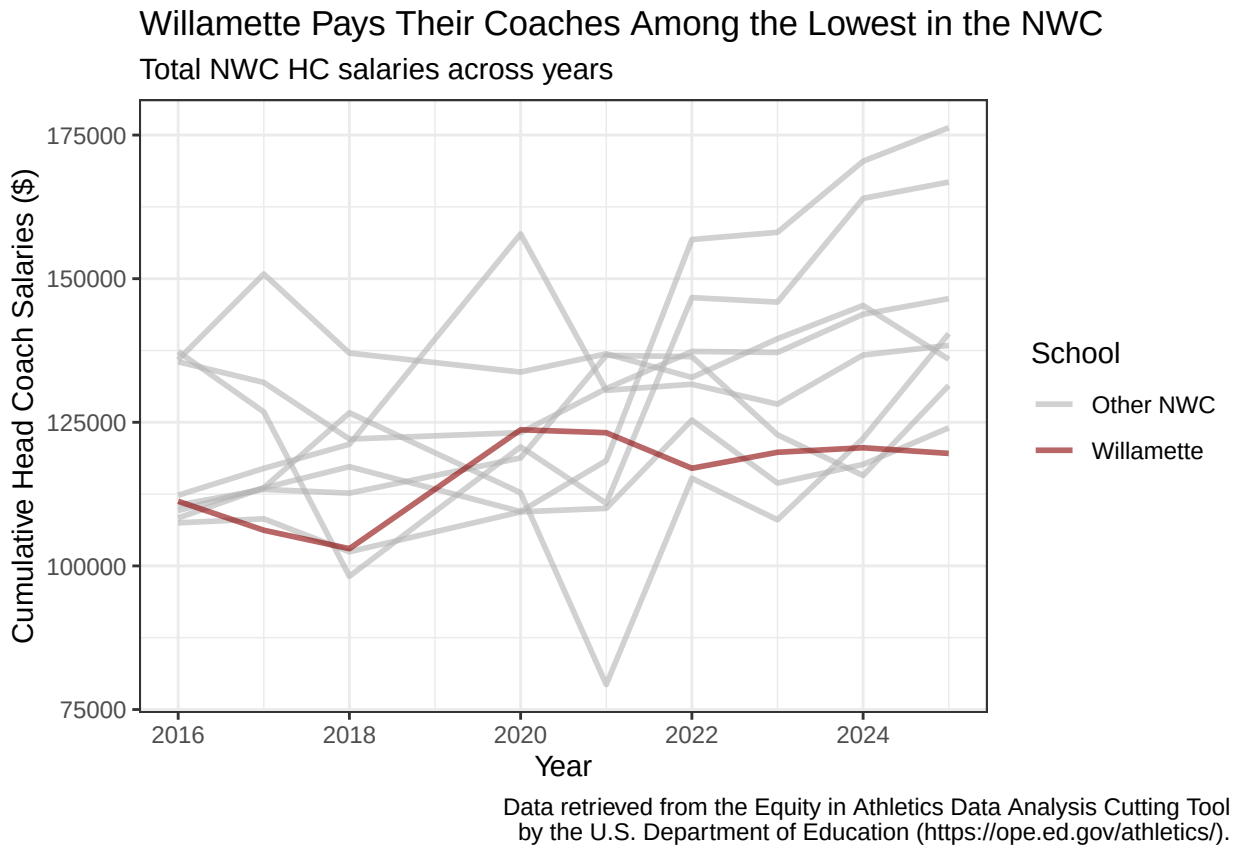
```
)


d3_sports %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University", year == 2025, sport %
  ggplot(aes(x = reorder_within(institution_name, total_hc + total_ac, sport), y = total_hc + total_ac)
  geom_col(aes(fill = institution_name == "Willamette University",
               group = institution_name)) +
  theme_light() +
  facet_wrap(
    ~ sport,
    scales = "free_x",
    labeller = as_labeller(c(
      "Women's Basketball" = "WBB",
      "Men's Basketball" = "MBB",
      "Women's Soccer" = "WSOC",
      "Men's Soccer" = "MSOC",
      "Women's Softball" = "Softball",
      "Men's Baseball" = "Baseball",
      "Men's Football" = "Football",
      "Women's Volleyball" = "Volleyball",
      "Men's Track and Field (Indoor)" = "M T&F (I)",
      "Men's Track and Field (Outdoor)" = "M T&F (O)",
      "Women's Track and Field (Indoor)" = "W T&F (I)",
      "Women's Track and Field (Outdoor)" = "W T&F (O)",
      "Women's Lacrosse" = "Lacrosse",
      "Men's Cross Country" = "M XC",
      "Women's Cross Country" = "W XC",
      "Men's Golf" = "M Golf",
      "Women's Golf" = "W Golf",
      "Men's Swimming" = "M Swim",
      "Women's Swimming" = "W Swim",
      "Women's Tennis" = "W Tennis",
      "Men's Tennis" = "M Tennis",
      "Women's Other Sports" = "W Other Sports"
    ))
  ) +
  scale_x_reordered(
    breaks = c()
  ) +
  scale_fill_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
  ) +
  labs(
    x = "",
    y = "Total Number of Coaches",
    fill = "School",
    title = "Is Willamette Undercoached?",
    subtitle = "Total # of coaches by sport in the NWC"
  )
```

Plot 4: Headcoach Salary vs Year

- Geometries: Lines
- Aesthetics: x = year, y = head coach salaries, color = type of school

```
d3_instLevel %>%
  mutate(HC_sal = HDCOACH_SAL_FTE_MEN+HDCOACH_SAL_FTE_WOMN) %>%
  filter(institution_name %in% nwc| institution_name == "Willamette University")%>%
  ggplot(aes(x = year, y = HC_sal)) +
  geom_line(linewidth = 1, alpha = .6, aes(color = institution_name == "Willamette University",
        group = institution_name)) +
  scale_x_continuous(breaks = c(2016,2018,2020,2022,2024))+
  scale_color_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
  )+
  theme_bw() +
  theme(
    plot.caption.position = "plot"
  ) +
  labs(
    title ="Willamette Pays Their Coaches Among the Lowest in the NWC",
    subtitle = "Total NWC HC salaries across years",
    y = "Cumulative Head Coach Salaries ($)",
    x = "Year",
    color = "School",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



Willamette, on average, per individual coach, ranks the lowest in 2025 in the Northwest Conference. This is significant because Willamette sports are continuing to generally struggle. Inability to pay top candidates hurts the ability of our teams to be the most competitive they can, because development of our athletes is limited. Investing higher amounts in coaches is needed to observe change instead of seeing a continuous slump.

```
d3_instLevel %>%
  mutate(HC_sal = HDCOACH_SAL_FTE_MEN+HDCOACH_SAL_FTE_WOMN) %>%
  filter(institution_name %in% nwc| institution_name == "Willamette University")%>%
  ggplot(aes(x = year, y = HC_sal, color = institution_name)) +
  geom_line(linewidth = 1, alpha = .6) +
  scale_x_continuous(breaks = c(2016,2018,2020,2022,2024))+
  # scale_color_manual(
  #   values = c("TRUE" = "darkred", "FALSE" = "gray70"),
  #   guide = "none"
  # )+
  theme_bw()


nwc_med = d3_instLevel %>%
  filter(institution_name %in% nwc) %>%
  summarize(
    .by = year,
    med_opexp_athlete = median(IL_TOTAL_OPEXP_INCLCOED/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN)),
    med_exp_athlete = median(GRND_TOTAL_EXPENSE/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))
  )
```

```
wu_med = d3_instLevel %>%
  filter(institution_name == "Willamette University") %>%
  summarize(
    .by = year,
    med_opexp_athlete = median(IL_TOTAL_OPEXP_INCLCOED/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN)),
    med_exp_athlete = median(GRND_TOTAL_EXPENSE/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))
    )


nwc_med %>%
  ggplot(aes(x = year)) +

  # Other NWC schools: operating expenses (dashed)
  geom_point(aes(y = med_opexp_athlete, color = "Other NWC Schools")) +
  geom_line(aes(y = med_opexp_athlete, color = "Other NWC Schools", linetype = "Operating Expenses")) +

  # Other NWC schools: total expenses (solid)
  geom_point(aes(y = med_exp_athlete, color = "Other NWC Schools")) +
  geom_line(aes(y = med_exp_athlete, color = "Other NWC Schools", linetype = "Total Expenses")
  ) +

  # Willamette: operating expenses (dashed)
  geom_point(data = wu_med, aes(y = med_opexp_athlete, color = "Willamette University")) +
  geom_line(data = wu_med, aes(y = med_opexp_athlete, color = "Willamette University", linetype = "Opera

  # Willamette: total expenses (solid)
  geom_point(data = wu_med, aes(y = med_exp_athlete, color = "Willamette University")) +
  geom_line(data = wu_med, aes(y = med_exp_athlete, color = "Willamette University", linetype = "Total

  scale_color_manual(
    name = "Institution",
    values = c(
      "Other NWC Schools" = "grey60",
      "Willamette University" = "darkred"
    )
  ) +
  scale_linetype_manual(
    name = "Expense Type",
    values = c(
      "Operating Expenses" = "dashed",
      "Total Expenses" = "solid"
    )
  ) +
  scale_x_continuous(
    breaks = seq(2000, 2025, by = 1)
  ) +
  theme_light() +
  labs(
    title = "Less of Willamette's Athletic Expenses Go Directly to Athletes",
    subtitle = "Compared with other NWC schools",
    x = "Year",
    y = "Expenses per Athlete ($)"
  )
```

```
nwc_ = d3_instLevel %>%
  filter(institution_name %in% nwc)

wu = d3_instLevel %>%
  filter(institution_name == "Willamette University")

nwc_ %>%
  bind_rows(wu) %>%
  ggplot(aes(x = year, y = RECRUITEXP_TOTAL/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))) +
  geom_point(aes(color = institution_name == "Willamette University")) +
  geom_smooth(data = wu, se = F, color = "darkred") +
  geom_smooth(data = nwc_, se = F, color = "grey70") +
  theme_light() +
  scale_color_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
  ) +
  scale_x_continuous(
    breaks = seq(2000, 2025, by = 1)
  ) +
  labs(
    title = "How Much is Spent on Recruiting Over Time",
    x = "Year",
    y = "Amount Spent on Recruiting per Athlete ($)",
    color = "Institution"
  )
```

Plot 5: Percent Difference in Expense Categories from NWC Median

- Geometries: Columns, vline
- Aesthetics: x = difference in percentage, y = expense category, fill = positive/negative

```
expense_breakdown = d3_instLevel %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University") %>%
  mutate(
    hc_salaries =
      coalesce(HDCOACH_SALARY_MEN * NUM_HDCOACH_MEN, 0) +
      coalesce(HDCOACH_SALARY_WOMEN * NUM_HDCOACH_WOMEN, 0) +
      coalesce(HDCOACH_SALARY_COED * NUM_HDCOACH_COED, 0),

    ac_salaries =
      coalesce(ASCOACH_SALARY_MEN * NUM_ASCOACH_MEN, 0) +
      coalesce(ASCOACH_SALARY_WOMEN * NUM_ASCOACH_WOMEN, 0) +
      coalesce(ASCOACH_SALARY_COED * NUM_ASCOACH_COED, 0),

    recruiting =
      coalesce(RECRUITEXP_MEN, 0) +
      coalesce(RECRUITEXP_WOMEN, 0) +
      coalesce(RECRUITEXP_COED, 0),

    operating = IL_TOTAL_OPEXP_INCLCOED,

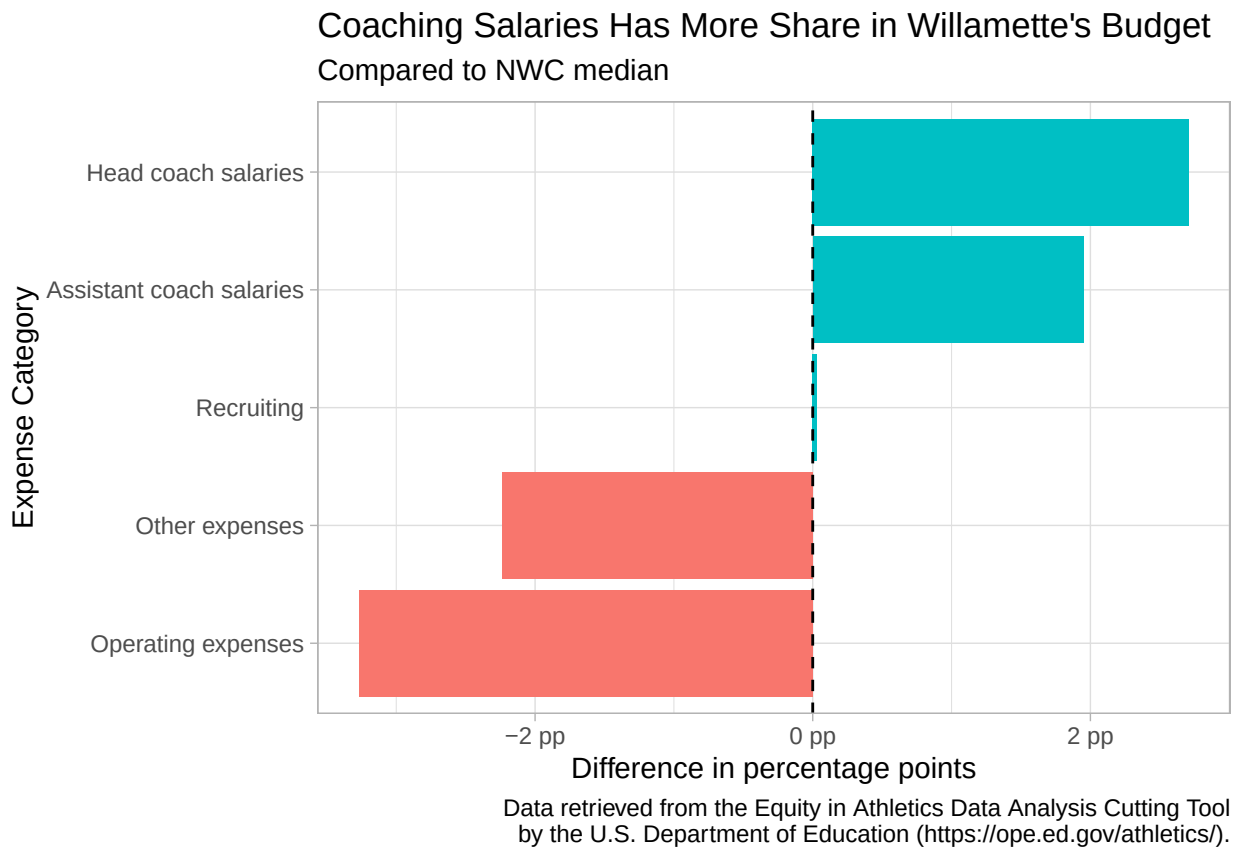
    other = GRND_TOTAL_EXPENSE - operating - recruiting - hc_salaries - ac_salaries
```

```
  )


selected_year = 2025
title = paste0("Where Willamette's Athletic Spending Differed in ", selected_year)

expense_breakdown %>%
  filter(year == selected_year) %>%
  mutate(
    operating = operating / GRND_TOTAL_EXPENSE * 100,
    recruiting = recruiting / GRND_TOTAL_EXPENSE * 100,
    hc_salaries = hc_salaries / GRND_TOTAL_EXPENSE * 100,
    ac_salaries = ac_salaries / GRND_TOTAL_EXPENSE * 100,
    other = other / GRND_TOTAL_EXPENSE * 100
  ) %>%
  select(
    institution_name,
    operating,
    recruiting,
    hc_salaries,
    ac_salaries,
    other
  ) %>%
  pivot_longer(
    cols = -institution_name,
    names_to = "category",
    values_to = "percentage"
  ) %>%
  group_by(category) %>%
  mutate(
    nwc_median = median(
      percentage[institution_name %in% nwc],
      na.rm = TRUE
    )
  ) %>%
  ungroup() %>%
  filter(institution_name == "Willamette University") %>%
  mutate(
    difference = percentage - nwc_median,
    category = recode(
      category,
      operating = "Operating expenses",
      recruiting = "Recruiting",
      hc_salaries = "Head coach salaries",
      ac_salaries = "Assistant coach salaries",
      other = "Other expenses"
    )
  ) %>%
  ggplot(aes(x = difference, y = reorder(category, difference), fill = difference > 0)) +
  geom_col() +
  geom_vline(xintercept = 0, linetype = "dashed") +
  scale_x_continuous(
    labels = \(x) paste0(x, " pp")
```

```
  ) +
  theme_light() +
  theme(
    legend.position = "none",
    plot.caption.position = "plot"
  ) +
  labs(
    title = "Coaching Salaries Has More Share in Willamette's Budget",
    subtitle = "Compared to NWC median",
    x = "Difference in percentage points",
    y = "Expense Category",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



While Willamette coaches need to be more invested in like our graph states above, the gap between expenses on athletes needs to be closed. As a result, more immediate change can be forecasted by increasing the percentage of our budget we spend on recruiting, operating expenses, and other expenses.

```
expense_breakdown %>%
  filter(year == 2025) %>%
  select(institution_name, operating, recruiting, hc_salaries, ac_salaries, other) %>%
  pivot_longer(
    cols = c(operating, recruiting, hc_salaries, ac_salaries, other),
    names_to = "category",
    values_to = "expense"
```

```
  ) %>%
  ggplot(aes(
    x = institution_name,
    y = expense,
    fill = category
  )) +
  geom_col() +
  labs(
    x = NULL,
    y = "Grand Total Expenses ($)",
    fill = "Expense Category"
  ) +
  theme_light() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )


expense_breakdown %>%
  filter(year == 2025) %>%
  select(
    institution_name,
    operating, recruiting,
    hc_salaries, ac_salaries, other
  ) %>%
  pivot_longer(
    cols = c(operating, recruiting, hc_salaries, ac_salaries, other),
    names_to = "category",
    values_to = "expense"
  ) %>%
  ggplot(aes(
    x = reorder(institution_name, expense),
    y = expense,
    fill = institution_name == "Willamette University"
  )) +
  geom_col() +
  facet_wrap(~ category, scales = "free_y") +
  coord_flip() +
  scale_fill_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    guide = "none"
  ) +
  labs(
    x = NULL,
    y = "Expenses ($)",
    title = "Athletic Expense Breakdown by School, 2025"
  ) +
  theme_light()



expense_breakdown %>%
  filter(year == 2025) %>%
  mutate(
```

```
    operating_pct = operating / GRND_TOTAL_EXPENSE * 100,
    recruiting_pct = recruiting / GRND_TOTAL_EXPENSE * 100,
    hc_salaries_pct = hc_salaries / GRND_TOTAL_EXPENSE * 100,
    ac_salaries_pct = ac_salaries / GRND_TOTAL_EXPENSE * 100,
    other_pct = other / GRND_TOTAL_EXPENSE * 100
  ) %>%
  select(
    institution_name,
    operating_pct, recruiting_pct,
    hc_salaries_pct, ac_salaries_pct, other_pct
  ) %>%
  pivot_longer(
    cols = ends_with("_pct"),
    names_to = "category",
    values_to = "percentage"
  ) %>%
  ggplot(aes(
    x = institution_name,
    y = percentage,
    fill = category
  )) +
  geom_col() +
  labs(
    x = NULL,
    y = "Percentage of Grand Total Expenses",
    fill = "Expense Category",
    title = "Athletic Expense Composition by School",
    subtitle = "2025"
  ) +
  scale_y_continuous(
    labels = \(x) paste0(x, "%"),
    limits = c(0, 100)
  ) +
  theme_light() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )


expense_breakdown %>%
  filter(year == 2025) %>%
  select(
    institution_name,
    operating, recruiting,
    hc_salaries, ac_salaries, other
  ) %>%
  pivot_longer(
    cols = -institution_name,
    names_to = "category",
    values_to = "percentage"
  ) %>%
  mutate(
    category = recode(
      category,
```

```
      operating = "Op Exp",
      recruiting = "Rec",
      hc_salaries = "HC Sal",
      ac_salaries = "AC Sal",
      other = "Other"
    )
  ) %>%
  group_by(institution_name) %>%
  mutate(
    category = reorder(category, percentage)
  ) %>%
  ungroup() %>%
  ggplot(aes(x = percentage, y = category)) +
  geom_col(fill = "gray65") +
  facet_wrap(~ institution_name, scales = "free_y") +
  labs(
    x = "Percent of Grand Total Expenses",
    y = NULL,
    title = "Athletic Expense Composition by School",
    subtitle = "2025"
  ) +
  theme_light()

expense_breakdown %>%
  mutate(
    operating_pct = operating / GRND_TOTAL_EXPENSE * 100
  ) %>%
  group_by(year) %>%
  mutate(
    nwc_median = median(operating_pct[institution_name %in% nwc], na.rm = TRUE)
  ) %>%
  ungroup() %>%
  filter(institution_name == "Willamette University") %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = operating_pct, color = "Willamette University"), linewidth = 1) +
  geom_line(aes(y = nwc_median, color = "Other NWC Schools"), linewidth = 1, linetype = "dashed") +
  labs(
    title = "Operating Expenses as a Share of Athletics Spending",
    x = "Year",
    y = "Operating expenses (% of total)",
    color = "Institution"
  ) +
  scale_color_manual(
    values = c(
      "Other NWC Schools" = "grey60",
      "Willamette University" = "darkred"
    )
  ) +
  scale_y_continuous(
    labels = \(x) paste0(x, "%")
  ) +
  scale_x_continuous(
    breaks = seq(2000, 2025, by = 1)
  ) +
```

```
theme_light()
```